THE LUNAR SURFACE
This Apollo 16 image is centered on the boundary between the near and far sides of the Moon—a view never seen before the era of spaceflight. At least 4 billion years of asteroid bombardment has saturated the lunar surface with craters.

MOON PROFILE

AVERAGE DISTANCE FROM EARTH 238,900 miles (384,400 km)	**ROTATION PERIOD** 27.32 Earth days
SURFACE TEMPERATURE –240ºF to 240ºF (–150ºC to 120ºC)	**LENGTH OF A DAY ON THE MOON** 29.53 Earth days
DIAMETER 2,160 miles (3,476 km)	**MASS (EARTH = 1)** 0.012
VOLUME (EARTH = 1) 0.02	**GRAVITY AT EQUATOR (EARTH = 1)** 0.165
NUMBER OF MOONS 0	**SIZE COMPARISON**

OBSERVATION
The amount of the sunlit Moon visible from Earth varies throughout the month, starting with a thin crescent in the western sky just after sunset. The month ends with a thin crescent Moon visible in the east just before dawn.

ATMOSPHERE

The Moon has a very thin, tenuous atmosphere with a total mass of about 22,000 lb (10,000 kg). This is the same as the amount of gas released by a landing Apollo spacecraft. The surface temperature varies by about 480°F (270°C) over a lunar day, and the quantity of gas near the surface is 20 times greater during the cold lunar night than during the heat of the day. The Moon's gravity is just one-sixth of Earth's, and the lunar atmosphere is escaping all the time. However, the atmosphere is also constantly being replenished by the solar wind.

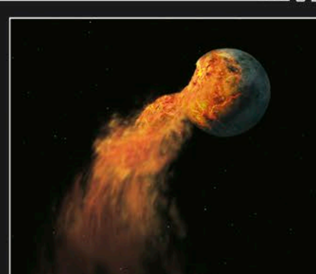
ATMOSPHERIC COMPOSITION
The neon, hydrogen, and helium have been captured from the solar wind. The argon is derived from the radioactive decay of potassium in the lunar rocks.

HISTORY OF THE MOON

No one knows exactly how the Moon was formed, but most astronomers agree with the giant-impact theory, which hypothesizes that the process was set in motion about 4.5 billion years ago, when a massive asteroid hit the young Earth (see below). During the first 750 million years of its life, the Moon went through a period of heavy meteorite bombardment, which cracked the crust and created craters all over the surface. About 3.5 billion years ago, the rate of bombardment slowed and there followed a period of considerable volcanic activity. Lava from 60 miles (100 km) below the surface oozed up through cracks in the crust and filled large, low-lying craters. The lava solidified, producing the dark, flat basaltic areas called maria. This volcanic activity stopped about 3.2 billion years ago, and since then, the Moon has been relatively dead. Many of the features formed in the early days of the Moon's history have been destroyed by subsequent impacts. One of the most recent large craters is Copernicus, which was produced about 900 million years ago.

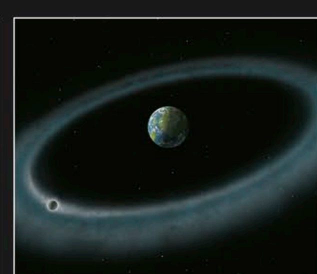
FORMATION OF EARTH'S MOON

1 In a glancing collision between a Mars-sized asteroid and Earth, a huge amount of silicate material was jetted away from Earth's mantle.



2 The ejected material formed a massive cloud of gas, dust, and rock. Heat was radiated away, and the cloud quickly began to cool.

3 The majority of the ejected material went into a circular orbit around Earth, forming a clumpy, dense, doughnut-shaped ring.



4 Rocks grew by mutual collisions until a single body dominated the ring, sweeping up the remaining material. The Moon was born.